How-To Library

Informatica

Enable Customer Managed Keys for MDM SaaS on Amazon Web Services

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Abstract

This article explains how to create a customer managed, master encryption key for the MDM SaaS solution on Amazon Web Services. You can control and maintain the key that you create. Use the key to control and restrict access to your organization's data. MDM SaaS leverages the keys that you control to encrypt data at rest, including data that persist in the data store, search indexes, and index backup.

Supported Versions

- Informatica MDM SaaS August 2023 or later

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Overview

Informatica Intelligent Cloud Services protects your organization's sensitive data in the cloud using organization-specific encryption keys that are generated and stored in the Informatica Intelligent Cloud Services key management service (KMS). To prevent malicious access, the keys are encrypted using a master key that is stored in the cloud provider's KMS.

If you prefer, you can create a customer-managed key (CMK). When you create a CMK, you control access to it. However, you'll need to grant Informatica Intelligent Cloud Services access to the CMK so that it can encrypt and decrypt your organization's sensitive data.

Creating a CMK offers the following benefits:

- You can restrict and control any access to your data.
- You can restrict the decryption of your data in emergency situations.
- You create and hold the key material in your KMS. The key is never exposed to your cloud service provider.
- You maintain full control of the key throughout its lifecycle. You can revoke access or delete the key at any time.

Note: When you create a CMK, your KMS and Informatica Intelligent Cloud Services POD must use the same cloud. For example, if your Informatica Intelligent Cloud Services POD is USW1 on AWS, then you must store your CMK in AWS KMS.

You can find instructions related to Amazon Web Services in this article for reference purposes. For more detailed information about Amazon Web Services, see the Amazon Web Services documentation.

Steps for creating and enabling the key

To create and use a CMK, you provision the key in AWS KMS and enable cross-account access with Informatica Intelligent Cloud Services. Then you enable customer managed keys in Informatica Intelligent Cloud Services.

To create a CMK, complete the following steps:

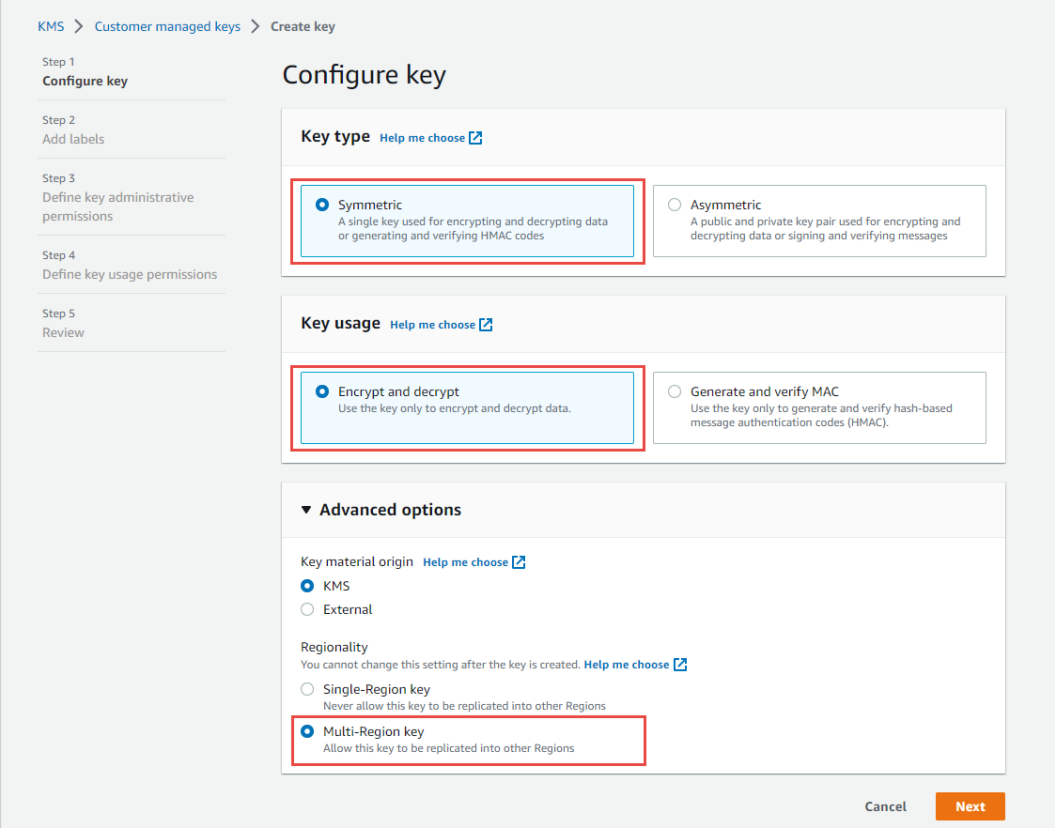
1. In AWS KMS, create the key to use as your CMK.
2. Create an IAM policy for the key.
3. Create an IAM role and attach the policy to the role so that Informatica can access the key.
4. Get the account ID and external ID from Informatica to create an IAM role for the MongoDB AWS account. Share ARN with Informatica so that Informatica can verify whether the database is configured for encryption.
5. Update the KMS policy for the IAM role created for the MongoDB AWS account.
6. In Informatica Intelligent Cloud Services Administrator, enable customer managed keys on the **Settings** page.

Step 1. Create the key in AWS KMS

Create a symmetric multi-regional key to use as your CMK with replicas in two regions as specified by Informatica Global Customer Support. Note the key ARN of the primary and replica keys because you'll need it when you enable customer managed keys in Informatica Intelligent Cloud Services.

1. Log in to the AWS Management Console.
2. In the search bar, search for **Key Management Service**.
3. Select **Customer managed keys**, and click **Create key**.
4. In **Key type**, select **Symmetric**.
5. In **Key usage**, select **Encrypt and decrypt**.

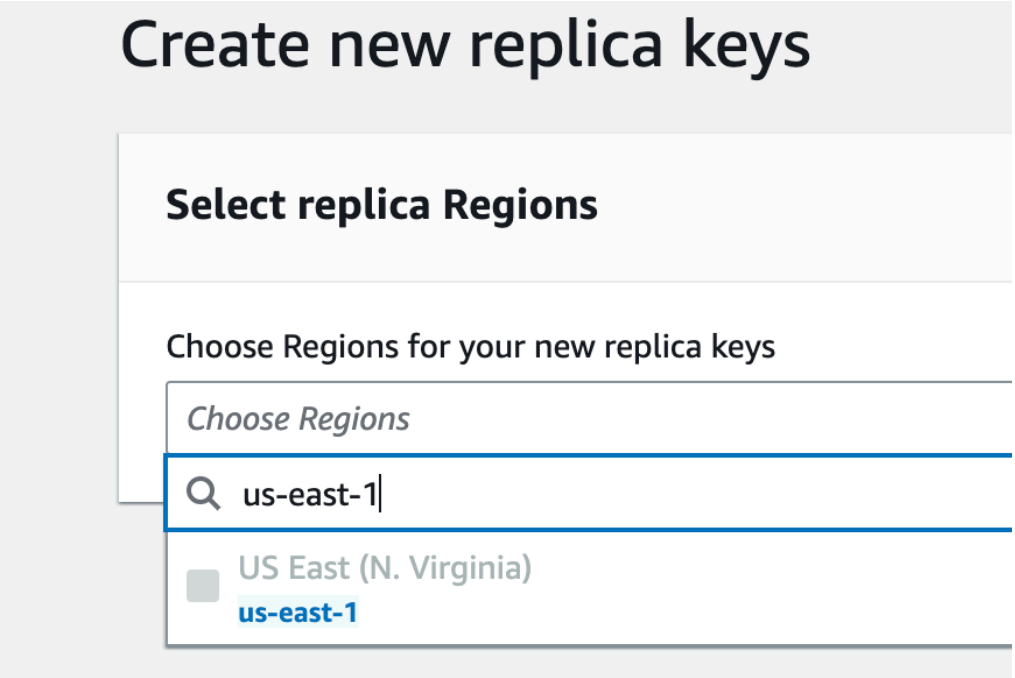
6. Expand the **Advanced options**, and select **Multi-Region key**.



7. Click **Next**.
8. Enter an alias for the key, and optionally add a description and tags.

   Tip: Enter an alias that indicates the type of data you plan to protect or the application you plan to use with the KMS key.
9. Select the IAM users and roles who can administer the key through the KMS API.
10. Click **Next**.
11. Select the IAM users and roles that can use the KMS key in cryptographic operations.
12. Click **Next**.
13. Review the key configuration and policy, and then click **Finish**.
14. To enable the key for disaster recovery in a different AWS region, perform the following tasks:
    a. Navigate to the key and select the **Regionality** tab.
    b. In the **Related multi-Region keys** section, select **Create new replica keys**.
    c. Select the disaster recovery region provided by Informatica.

The following image shows the us-east-1 region as the disaster recovery region:



d. Keep the key policy the same, and select **I understand that the values I choose here are not synchronized with any other multi-Region key** and click **Create new replica keys**.

15. Note the KMS key ARN of the primary and replica keys and share it with Informatica.

Step 2. Create an IAM policy for the key

In the AWS Management Console, create an IAM policy for the CMK to define its permissions. Note the policy name because you'll need it when you create an IAM role to access the key.

1. In the search bar, search for **IAM**.
2. Under **Access Management**, navigate to **Policies**.
3. Click **Create Policy**, switch to the JSON view, and enter the following text:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "VisualEditor0",
      "Effect": "Allow",
      "Action": [
        "kms:Decrypt",
        "kms:Encrypt",
        "kms:DescribeKey"
        "kms:ListKeyRotations",
        "kms:ListKeys"
      ],
      "Resource": "{your key ARN}"
    }
  ]
}
```

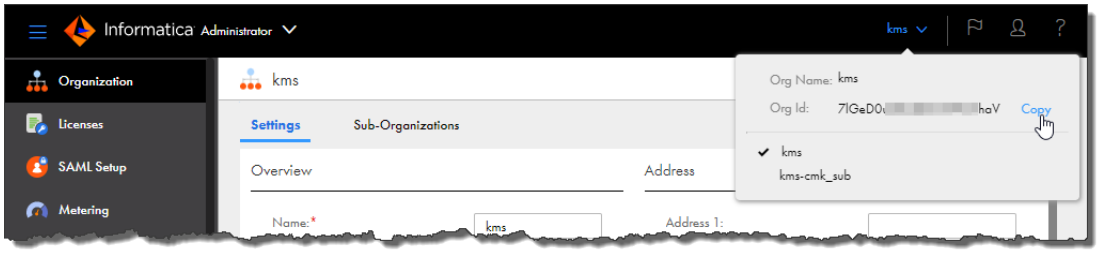
4. Click **Next**.
5. Optionally, add tags to the policy.
6. Click **Next**.

7. Enter a meaningful name for the policy, for example, `Informatica-KMS-Access`, and click **Create policy**.

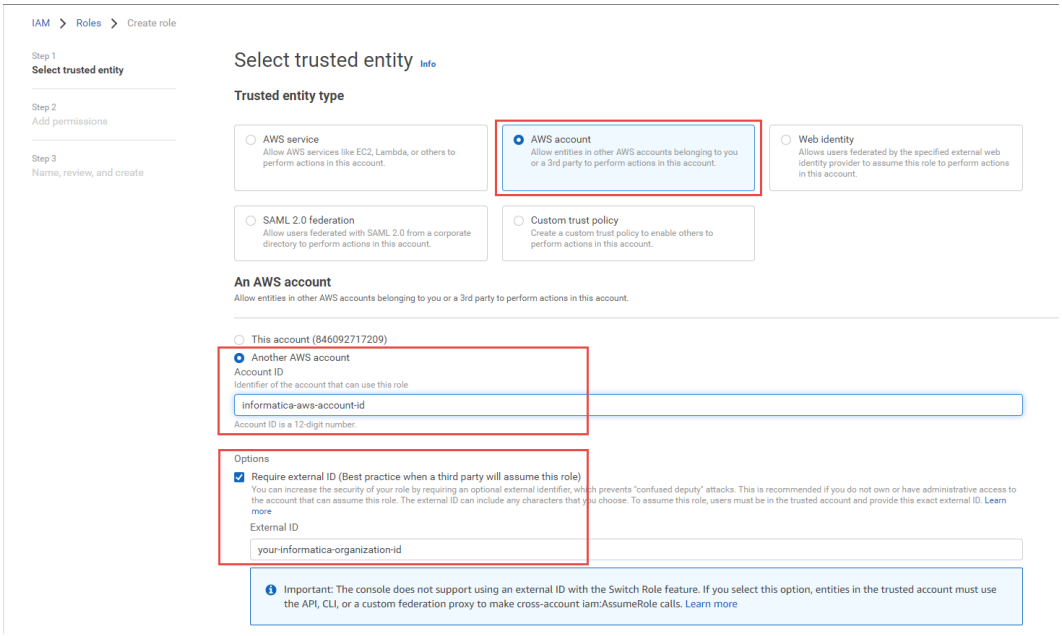
Step 3. Create an IAM role to access the key

In the AWS Management Console, create an IAM role to allow Informatica Intelligent Cloud Services to access your CMK. Note the role ARN because you'll need it when you enable customer managed keys in Informatica Intelligent Cloud Services.

To complete this step, you'll need your Informatica Intelligent Cloud Services organization ID. You can find your organization ID in the upper right corner when you log in to any service in Informatica Intelligent Cloud Services:



1. In the search bar, search for **IAM**.
2. Under **Access Management**, navigate to **Roles** and click **Create role**.
3. Set the trusted entity type to **AWS account**.
4. In the **Account ID** field, enter Informatica's AWS account ID: `130917795281`
5. Enable **Require external ID**, and enter your Informatica Intelligent Cloud Services organization ID as the external ID:



6. Click **Next**.
7. Search for the policy that you created in "Step 2. Create an IAM policy for the key" on page 5, for example, `Informatica-KMS-Access`, and select it.
8. Click **Next**.

9. Enter a name for the role.

   Important: The role name must use the following prefix: `informatica-kms-cmk-access-role-`

   For example, you might name the role, `informatica-kms-cmk-access-role-prod`.
10. Review the role permissions and click **Create role**.
11. On the **Roles** page, select the role you created and note the role ARN.
12. Provide the role ARN to Informatica Global Customer Support.

Step 4. Create an IAM role for the MongoDB AWS account

In the AWS Management Console, create an IAM role for the MongoDB AWS account to allow Informatica Intelligent Cloud Services to access your CMK. Contact Informatica Global Customer Support to get the external MongoDB AWS account and the external ID details. Perform this task with help from Informatica Global Customer Support.

To access the MongoDB AWS account and external ID, contact Informatica Global Customer Support.

1. In the search bar, search for **IAM**.
2. Under **Access Management**, navigate to **Roles** and click **Create role**.
3. Set the trusted entity type to **AWS account**.
4. In the **Account ID** field, enter the account ID provided by Informatica Global Customer Support.
5. Enable **Require external ID**, and enter the external ID provided by Informatica Global Customer Support.
6. Click **Next**.
7. To add the custom trust relationship policy, enter the following text:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "<atlasAWSAccountArn>"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "sts:ExternalId": "<atlasAssumedRoleExternalId>"
        }
      }
    }
  ]
}
```

8. Click **Next**.
9. Enter a name for the role.
10. Review the role permissions and click **Create role**.
11. On the **Roles** page, select the role you created and note the role ARN.
12. Provide the role ARN to Informatica Global Customer Support.

    Informatica Global Customer Support adds the ARN and continues with the subsequent steps to verify whether the database is configured for encryption.

Step 5. Update the KMS key policy

You'll need to add the IAM roles to the KMS key policy and update the KMS key policy for the replica key.

Update the KMS key policy for the roles created in "Step 4. Create an IAM role for the MongoDB AWS account" on page 7. You need to obtain the Informatica AWS Account number and IAM roles from Informatica Global Customer Support before updating the policy.

1. In the search bar, search for **IAM**.
2. Under **Access Management**, navigate to **Policies**.
3. In the **Key Policy** section, choose **Switch to policy view**.
4. Add the following policy to the KMS key:

```
{
    "Version": "2012-10-17",
    "Id": "key-consolepolicy",
    "Statement": [
        {
            "Sid": "Enable IAM User Permissions",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::<Your AWS Account ID>:root"
            },
        "Statement": [
        {
            "Sid": "Enable IAM User Permissions",
            "Effect": "Allow",
            "Principal": {
                "AWS": "arn:aws:iam::<Your AWS Account ID:root"
            },
            "Action": "kms:*",
            "Resource": "*"
        }
        {
            "Sid": "Assume Role for Informatca & MongoDB",
            "Effect": "Allow",
            "Principal": {
                "AWS": [
                    "arn:aws:iam::<MongoDB Assume IAM Role ARN Created by You>",
                    "arn:aws:iam::<Informatica Assume IAM role ARN Created by You",
                ]
            },
            "Action": [
                "kms:Encrypt",
                "kms:DescribeKey",
                "kms:Decrypt",
                "kms:ReEncrypt*",
                "kms:CreateGrant"
            ],
            "Resource": "<YOUR PRIMARY KMS KEY ARN>"
        },
        {
        "Sid": "Allow external account <Informaticas AWS account number> use of the
        customer managed key",
        "Effect": "Allow",
        "Principal": {
            "AWS": [
                "<IAM ROLE 1 ARN PROVIDED BY INFORMATICA>",
                "<IAM ROLE 2 ARN PROVIDED BY INFORMATICA>"
            ]
        },
            "Action": [
                "kms:Encrypt",
                "kms:Decrypt",
                "kms:ReEncrypt*",
                "kms:GenerateDataKey*",
```

```
                "kms:DescribeKey",
                "kms:CreateGrant"
            ],
            "Resource": "<YOUR PRIMARY KMS KEY ARN>"
        }
    ]
}
```

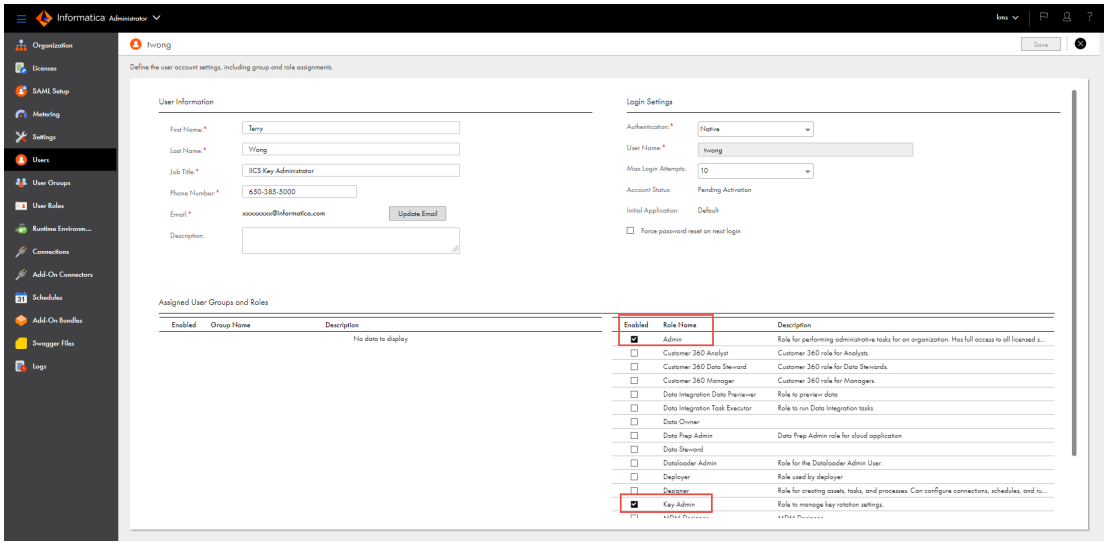
5. Click **Save changes**.
6. To update the KMS key policy in the replica key, perform the following steps:
   a. Navigate to the key and select the **Regionality** tab.
   b. Click on the replica key created for the disaster recovery region.
   c. Click **Key Policy** and click **Edit**.
   d. Update your primary KMS key ARN used in the primary key policy to use as the replica KMS key ARN for the disaster recovery region. You'll need to update at three different locations with the replica key.

      Note: The Key ID remains the same, but the region in the ARN will differ. You can locate the exact Key ARN at the top of the page under **General Configuration**.
   e. Click **Save changes**.

Step 6. Enable customer managed keys in Informatica Intelligent Cloud Services

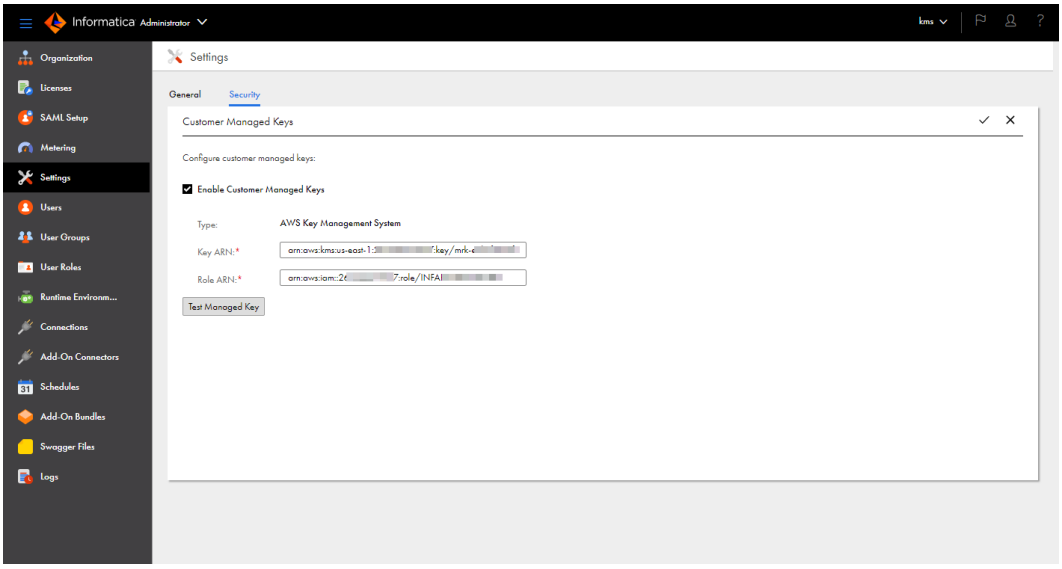
In Informatica Intelligent Cloud Services Administrator, open the **Settings** page and enable customer managed keys for your organization.

Note: Before you can complete this step, you need to assign at least one administrative user the **Admin** and **Key Admin** roles on the user details page in Administrator:



1. Log in to Informatica Intelligent Cloud Services Administrator with a user account that has both the Admin and Key Admin roles.
2. Open the **Settings** page and click the **Security** tab.
3. Click the edit (pencil) icon.
4. Enable the **Enable Customer Managed Keys** option.

5. Enter the Key ARN that you created in "Step 1. Create the key in AWS KMS" on page 3 and the Role ARN that you created in "Step 3. Create an IAM role to access the key" on page 6:



6. Click **Test Managed Key** to test the key.

   A success message appears if the test was successful.

7. Click the save (checkmark) icon to save your changes.

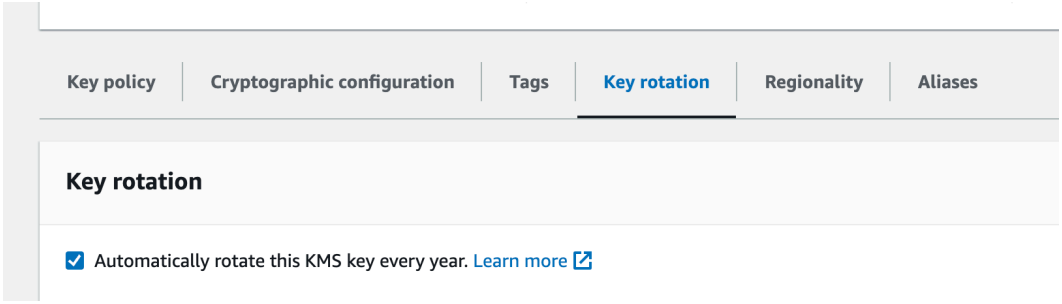
   Note: It can take up to 24 hours for the key to become active.

Enabling key rotation

You can enable or disable the automatic key rotation for multi-Region keys. You can manage automatic rotation only on the primary key. Automatic key rotation is a shared property of multi-Region keys. When you enable automatic key rotation, AWS KMS rotates the KMS key every year.

Note: Informatica does not currently support manual key rotation.

1. To enable key rotation, log in to the AWS Management Console.
2. Click the **Key rotation** tab and select **Automatically rotate this KMS key every year**.



3. Click **Save**.

Key revocation

To restrict Informatica's access to your data temporarily, you can disable the key in AWS KMS. After you disable the CMK, Informatica Cloud Services can't decrypt your organization's encrypted data, and any processes or jobs relying on this data will fail until you reactivate the CMK in AWS KMS.

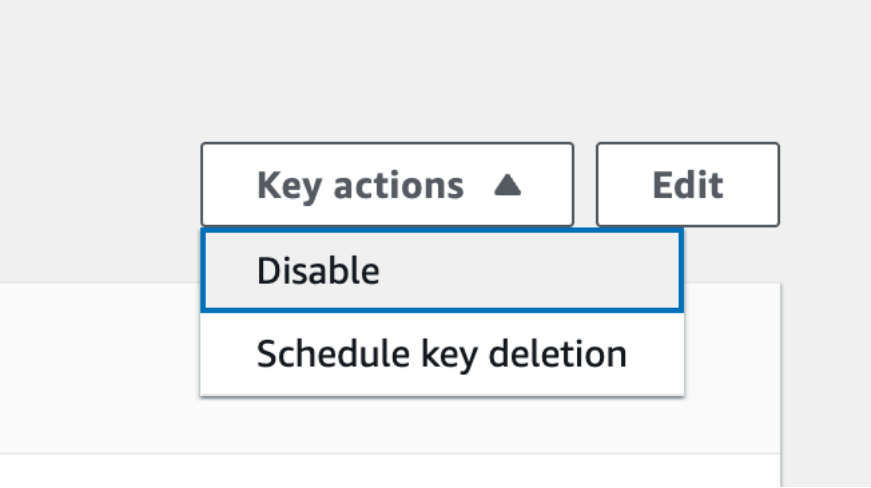
To revoke access to Elasticsearch data, contact Informatica Global Customer Support. The search indexes are stored on Amazon Elastic Block Store (Amazon EBS), and the Amazon EBS volume retains the key in memory, which gets removed upon server reboot. The Informatica operations team manually shuts down the Elasticsearch cluster. Any subsequent operations fail if the key remains disabled.

Enabling and disabling KMS keys

You can use the AWS KMS console to enable and disable customer managed keys.

1. Log in to the AWS Key Management Service (AWS KMS) console.
2. To change the AWS region, use the region selector in the upper-right corner of the page.
3. In the navigation pane, select **Customer managed keys**.
4. To enable a KMS key, select **Enable** from the **Key actions** list.

   To disable a KMS key, select **Disable** from the **Key actions** list.



Note: If you delete the CMK in AWS KMS, you permanently revoke Informatica's access to your encrypted data in Informatica Intelligent Cloud Services. In such a scenario, Informatica can't retrieve the data.

Frequently asked questions

What happens if the CMK is rotated in AWS KMS?

Informatica only supports automatic key rotation provided by AWS, and manual key rotation is not supported.

What if I want to temporarily revoke Informatica's access to the CMK?

If you want to temporarily revoke Informatica's access to the CMK, you can disable the key in AWS KMS.

When you disable the CMK, Informatica Intelligent Cloud Services can't decrypt your organization's encrypted data, and any jobs that use the data fails until you reactivate the CMK in AWS KMS. Disabling the key doesn't impact the Elasticsearch service cluster because AWS EBS keeps the key in memory. To disable the cluster,

contact Informatica Global Customer Support. Informatica will manually shut it down. Any subsequent operations fail if the key remains disabled.

If you permanently revoke Informatica's access to your encrypted data in Informatica Intelligent Cloud Services, you can delete the CMK in AWS KMS. In this event, Informatica can't recover the data.

Can I use separate encryption keys for Development, Production, and QA organizations?

Informatica requires you to use a single master key for the deployed organizations. MDM SaaS leverages the master key to encrypt unique tenant-specific keys, ensuring that data isolation and security are maintained for each tenant individually.

Can I request my organizations to be deployed on different PODs?

When you use Customer Managed Keys, all organizations must be deployed within a single POD.

Author

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